APPENDIX D

Density Matrices

D.1 Normal Densities

For the description of the dynamics of a many-body system, we often use *density matrices*. We distinguish between one-particle density matrices, two-body density matrices, and so on. In this book we use mostly one-particle densities. For higher densities the reader is referred to the work of [Th 61b]. We shall also restrict ourselves to the fermion case.

First we define a single-particle operator $\hat{\rho}(\mathbf{r})$ in an N-body Hilbert space:

$$\hat{\rho}(\mathbf{r}) = \sum_{i=1}^{N} \delta(\mathbf{r} - \hat{\mathbf{r}}_i), \tag{D.1}$$

where $\hat{\mathbf{r}}_i$ is the space operator of particle i and $\mathbf{r}$ is a parameter. $\hat{\rho}(\mathbf{r})$ may be expressed in the framework of second quantization (see Appendix C):

$$\hat{\rho}(\mathbf{r}) = \sum_{pq} d_{pq} a_p^+ a_q, \tag{D.2}$$

$$d_{pq} = \langle p | \delta(\mathbf{r} - \hat{\mathbf{r}}) | q \rangle = \sum_s \varphi_p^*(\mathbf{r}, s) \varphi_q(\mathbf{r}, s), \tag{D.3}$$

$$\hat{\rho}(\mathbf{r}) = \sum_s a^+(\mathbf{r}, s) a(\mathbf{r}, s). \tag{D.4}$$

The expectation value of this operator in an N-body state $|\Psi\rangle$ is (see

Appendix C)

$$\langle\Psi|\hat{\rho}(\mathbf{r})|\Psi\rangle = \sum_s \langle\Psi|a^+(\mathbf{r},s)a(\mathbf{r},s)|\Psi\rangle \tag{D.5}$$

$$= N \sum_{ss_2\ldots s_N} \int d^3r_2 \ldots d^3r_N |\Psi(\mathbf{r},s,\mathbf{r}_2,s_2,\ldots,\mathbf{r}_N,s_N)|^2 = \rho(\mathbf{r}), \tag{D.6}$$

which expresses the fact that the expectation value of $\hat{\rho}(\mathbf{r})$ is just the average particle density $\rho(\mathbf{r})$ at the point $\mathbf{r}$ in this state. Integrating (D.6) over the whole space gives the particle number N of the system.

Equation (D.6) can also be interpreted as the diagonal element of an operator $\hat{\rho}_\Psi$ in the coordinate space representation, which is called the *density matrix*. In general, it is defined as

$$\langle \mathbf{r},s|\hat{\rho}_\Psi|\mathbf{r}',s'\rangle := \rho(\mathbf{r}s;\mathbf{r}'s') = \langle\Psi|a^+(\mathbf{r}',s')a(\mathbf{r},s)|\Psi\rangle. \tag{D.7}$$

From Eq. (C.26) we get

$$\rho(\mathbf{r},s;\mathbf{r}',s') = \sum_{pq} \varphi_p(\mathbf{r},s)\rho_{pq}\varphi_q^*(\mathbf{r}',s')$$

$$= \sum_{pq} \langle \mathbf{r},s|p\rangle \rho_{pq} \langle q|\mathbf{r}',s'\rangle, \tag{D.8}$$

where

$$\rho_{pq} = \langle\Psi|c_q^+ c_p|\Psi\rangle \tag{D.9}$$

is the matrix element of the density operator $\hat{\rho}_\Psi$ in an arbitrary basis and $\hat{\rho}_\Psi$ has the form

$$\hat{\rho}_\Psi = \sum_{pq} |p\rangle \rho_{pq} \langle q|. \tag{D.10}$$

It is a Hermitian single-particle operator. The matrix ρ can be diagonalized by a unitary transformation of the single-particle basis

$$(D^+ \rho D)_{ll'} = \rho_l \delta_{ll'}, \qquad a_l^+ = \sum_{l'} D_{l'l} c_{l'}^+, \tag{D.11}$$

where

$$\rho_l = \langle\Psi|a_l^+ a_l|\Psi\rangle, \qquad 0 \leqslant \rho_l \leqslant 1, \tag{D.12}$$

is the probability that the level l is occupied in the wave function Ψ. Here we wrote ρ down in occupation number representation (see Appendix C) in the basis where ρ is diagonal. For the particle number we get

$$N = \sum_l \rho_l = \mathrm{Tr}\,\rho = \sum_s \int \rho(\mathbf{r},s;\mathbf{r},s)\, d^3r = \int \rho(\mathbf{r})\, d^3r \tag{D.13}$$

and for any other single particle operator of the form (C.32) we get

$$\langle\Psi|\hat{F}|\Psi\rangle = \sum_{ll'} f_{ll'} \langle\Psi|a_l^+ a_{l'}|\Psi\rangle = \mathrm{Tr}(f\cdot\rho). \tag{D.14}$$

There are two important examples of single-particle densities, which we will treat in some detail.

D.2 Densities of Slater Determinants

In the case of product wave functions we have the following *theorem*.

A wave function $\Psi(1\ldots N)$ *is a Slater determinant if and only if the corresponding density matrix* ρ_Ψ (D.10) *is a projector in the single-particle Hilbert space, that is,*

$$\hat{\rho}_\Psi^2 = \hat{\rho}_\Psi. \tag{D.15}$$

There is a one-to-one correspondence between $\hat{\rho}_\Psi$ *and* $|\Psi\rangle$ *in this case (of course, only up to a phase factor in* $|\Psi\rangle$*).*

To prove this theorem we start with a Slater determinant

$$|\Psi\rangle = a_1^+ \ldots a_N^+ |-\rangle \tag{D.16}$$

in some single-particle basis characterized by the operators a_l^+, a_l. From (D.9) we see that ρ is diagonal in this basis, with

$$\rho_l = \begin{cases} 0 & \text{for } l > N \quad \text{(empty levels, particles),} \\ 1 & \text{for } l \leqslant N \quad \text{(occupied levels, holes).} \end{cases} \tag{D.17}$$

$\hat{\rho}_\Psi$ (D.10) has the form

$$\hat{\rho}_\Psi = \sum_{i=1}^{N} |i\rangle\langle i|. \tag{D.18}$$

This is a projector onto the space of occupied states with the property (D.15).

On the other hand, we can start with a density matrix ρ with the property (D.15). It can be diagonalized [see Eq. (D.11)] and the eigenvalues ρ_l have the property

$$\rho_l^2 = \rho_l, \tag{D.19}$$

that is, they are either 0 or 1. Constructing a Slater determinant from the single-particle wave functions with $\rho_l = 1$ gives us the corresponding wave function Ψ. These single-particle wave functions φ_i are not uniquely determined by the diagonalization of ρ. Any unitary transformation among the occupied levels leaves ρ invariant. As we see from (D.16), however, such a unitary transformation multiplies the wave function $|\Psi\rangle$ only by the determinant of this transformation, which is a phase.

This shows that there is a one-to-one correspondence between ρ and $|\Psi\rangle$ in the case of Slater determinants. In particular, we can use Wick's theorem (C.52) and express the expectation value $\langle\Psi|\hat{O}|\Psi\rangle$ of an arbitrary operator $\hat{O}$ by the single-particle density ρ.

We next prove a *theorem due to Baranger and Veneroni* [BV 78, RS 77b], which states that any single-particle density matrix ρ that belongs to a Slater determinant ($\rho^2 = \rho$) can be decomposed in the following way.

$$\rho = e^{ix} \rho_0 e^{-ix}, \tag{D.20}$$

where χ and ρ_0 are Hermitian matrices that are even under time reversal. This decomposition is unique if we require

(i) that χ has only *ph* and *hp* matrix elements in the basis in which ρ_0 is diagonal:

$$\rho_0\chi\rho_0=\sigma_0\chi\sigma_0=0 \tag{D.21}$$

($\sigma_0=1-\rho_0$ projects onto particle states);

(ii) that the eigenvalues χ_μ of the matrix χ have the property

$$-\frac{\pi}{4}\leqslant\chi_\mu<\frac{\pi}{4}. \tag{D.22}$$

To prove this and to establish the uniqueness of the decomposition (D.20) we use the operator

$$\tau=2\rho-1 \qquad \text{with} \quad \tau^2=1. \tag{D.23}$$

The product, with its time reserved operator $\tau_T=T\tau T^+$, is unitary and is therefore of the form

$$\tau\tau_T=e^{4i\chi}, \tag{D.24}$$

where the Hermitian operator χ is uniquely defined by the condition (D.22). By inversion, time reversal, and Hermitian conjugation,

$$\tau_T\tau=e^{-4i\chi}=e^{-4i\chi_T}=e^{-4i\chi^+}, \tag{D.25}$$

we see that χ is time-reversal invariant and Hermitian.

Let $|\mu\rangle$ be a set of eigenstates of $\tau\tau_T$ with eigenvalue $\mu=e^{4i\chi_\mu}$, then $T|\mu\rangle$ is another eigenvector with the same eigenvalue μ and $\tau|\mu\rangle$ is an eigenvector with eigenvalue μ^*. This implies

$$(\chi\tau+\tau\chi)|\mu\rangle=(-\chi_\mu+\chi_\mu)\tau|\mu\rangle=0, \tag{D.26}$$

and thus

$$\chi\tau+\tau\chi=0. \tag{D.27}$$

This means, in particular, that for an arbitrary real number α,

$$e^{i\alpha\chi}\tau=\tau e^{-i\alpha\chi} \tag{D.28}$$

and

$$\tau_0:=e^{-2i\chi_T}=\tau e^{2i\chi}=\tau e^{-2i\chi}\tau\tau_T=e^{2i\chi}\tau_T=\tau_T e^{-2i\chi}.$$

τ_0 is therefore Hermitian and time even. The same holds for

$$\rho_0:=\tfrac{1}{2}(\tau_0+1). \tag{D.29}$$

Using Eq. (D.28) again gives

$$\chi\rho_0+\rho_0\chi=\chi, \tag{D.30}$$

which is equivalent to Eq. (D.21). Q.E.D.

In the following, we give *some rules* for calculating with single-particle densities ρ of Slater determinants ($\rho^2=\rho, \sigma=1-\rho$). An arbitrary matrix A

has the following *pp*, *ph*, *hp*, and *hh* parts in a basis in which ρ is diagonal:

$$A^{pp} := \sigma A \sigma; \qquad A^{hh} = \rho A \rho; \qquad A^{ph} = \sigma A \rho; \qquad A^{hp} = \rho A \sigma. \quad \text{(D.31)}$$

The three statements

$$A = A\rho + \rho A \Leftrightarrow A = \sigma A + A\sigma \Leftrightarrow A^{pp} = A^{hh} = 0 \quad \text{(D.32)}$$

are equivalent. If two matrices A and B obey the relation $B = [A, \rho]$ it follows:

$$B^{pp} = B^{hh} = 0; \qquad B^{ph} = A^{ph}; \qquad B^{hp} = -A^{hp}. \quad \text{(D.32)}$$

If, in addition,

$$A^{pp} = A^{hh} = 0$$

we can write

$$A = [B, \rho]. \quad \text{(D.33)}$$

For Hermitian matrices A, B with vanishing *pp* and *hh* matrix elements, we often define vectors

$$\begin{pmatrix} A \\ A^* \end{pmatrix} = \begin{pmatrix} A_{mi} \\ A^*_{mi} \end{pmatrix}, \quad \text{(D.34)}$$

and find the relations

$$\begin{aligned} (A^* A)\begin{pmatrix} B \\ B^* \end{pmatrix} &= \sum_{mi} A^*_{mi} B_{mi} + A_{mi} B^*_{mi} = \mathrm{Tr}(A \cdot B) \\ (A^* A)\begin{pmatrix} B \\ -B^* \end{pmatrix} &= \mathrm{Tr}(A \cdot [B, \rho]). \end{aligned} \quad \text{(D.35)}$$

Next we investigate some properties of a family of Slater determinants $|\Phi(q)\rangle$ depending on some parameter q (e.g. the deformation) with the densities $\rho(q)$. Starting from one wave function in this family, $|\Phi(q_0)\rangle$, we can represent the other wave functions by a Hermitian single-particle operator $\hat{P}$, which has only *ph* and *hp* matrix elements with respect to $|\Phi(q_0)\rangle$, as follows [see Eq. (E.40)].

$$|\Phi(q + q_0)\rangle = e^{-(i/\hbar) q \hat{P}} |\Phi(q_0)\rangle, \quad \text{(D.36)}$$

$$\hat{P} = P^0 + \sum_{mi} P_{mi} a_m^+ a_i + P^*_{mi} a_i^+ a_m. \quad \text{(D.37)}$$

The constant P^0 eventually determines a phase. The operator $\hat{P}$ generally depends on q and q_0, and only in the limit $q \to 0$ does it become independent of q and we then gain

$$\hat{P} |\Phi(q)\rangle = i\hbar \frac{\partial}{\partial q} |\Phi(q)\rangle. \quad \text{(D.38)}$$

For the density we have

$$\rho(q_0 + q) = e^{-(i/\hbar) q P} \rho(q_0) e^{(i/\hbar) q P} \quad \text{(D.39)}$$

and

$$\frac{\partial}{\partial q}\rho(q)=\frac{-i}{\hbar}[P,\rho]. \tag{D.40}$$

From the representation (D.18) we have

$$\frac{\partial}{\partial q}\hat{\rho}_{\Psi}=\frac{\partial}{\partial q}\sum_i |i\rangle\langle i|. \tag{D.41}$$

The *ph*-elements of $\hat{P}$ therefore take the form

$$P_{mi}=i\hbar\langle m|\frac{\partial}{\partial q}|i\rangle=i\hbar\int d\mathbf{r}\,\varphi_m^*(\mathbf{r},q)\frac{\partial}{\partial q}\varphi_i(\mathbf{r},q). \tag{D.42}$$

Very often the $|\Phi(q)\rangle$ are determined as HF solutions (see Sec. 7.6) of a Hamiltonian $H(q)$, which depends on a parameter q. From the variational principle (5.5) we get the Feynman theorem [Fe 39]:

$$\frac{d}{dq}\langle\Phi(q)|H(q)|\Phi(q)\rangle=\langle\Phi(q)|\frac{\partial H}{\partial q}|\Phi(q)\rangle. \tag{D.43}$$

In cases in which $\partial H/\partial q$ is a single-particle operator, we can calculate the matrix elements P_{mi} in linear response theory (see Sec. 8.5.3). If we neglect the residual interaction, a simple perturbation theory gives (see, for instance, Sec. 3.4.2)

$$\langle m|\frac{\partial}{\partial q}|i\rangle=\frac{\langle m|\partial H/\partial q|i\rangle}{\epsilon_m-\epsilon_i}, \tag{D.44}$$

where ϵ_m, ϵ_i are the single-particle energies corresponding to the operator $H(q)$.

D.3 Densities of BCS and HFB States

In the case of generalized product wave functions $|\Phi\rangle$ (see Sec. 7.2.3), we had defined the generalized density matrix

$$\mathcal{R}=\begin{pmatrix}\langle\Phi|c_{l'}^+c_l|\Phi\rangle & \langle\Phi|c_{l'}c_l|\Phi\rangle\\ \langle\Phi|c_{l'}^+c_l^+|\Phi\rangle & \langle\Phi|c_{l'}c_l^+|\Phi\rangle\end{pmatrix}=\begin{pmatrix}\rho_{ll'} & \kappa_{ll'}\\ -\kappa_{ll'}^* & 1-\rho_{ll'}^*\end{pmatrix}. \tag{D.45}$$

Besides the normal density ρ (D.9), it contains the anomalous density κ (which is also called the pairing tensor). Under a unitary transformation D of the basis [for instance, Eq. (D.11)], the matrices ρ and κ transform in the following way.

$$\rho\to D^+\rho D, \qquad \kappa\to D^+\kappa D^*. \tag{D.46}$$

We can therefore represent κ in coordinate space [see Eq. (C.26)]:

$$\kappa(\mathbf{r},s;\mathbf{r}',s')=\langle\Phi|a(\mathbf{r}',s')a(\mathbf{r},s)|\Phi\rangle=\sum_{pq}\varphi_p(\mathbf{r},s)\kappa_{pq}\varphi_q(\mathbf{r}',s'). \tag{D.47}$$

In the canonical basis $|k\rangle$ (see Sec. 7.2.1) or in the pure BCS case, κ_{pq} has

canonical form (7.25) and we get

$$\kappa(\mathbf{r},s;\mathbf{r}',s')=\sum_{k>0}(\varphi_k(\mathbf{r},s)\varphi_{\bar{k}}(\mathbf{r}',s')-\varphi_{\bar{k}}(\mathbf{r},s)\varphi_k(\mathbf{r}',s'))u_k v_k. \quad \text{(D.48)}$$

Working with the generalized density $\mathcal{R}$, we can show that all the results we have obtained for Slater determinants also apply to this case.

D.4 The Wigner Transformation of the Density Matrix

The Wigner transformation of the density matrix is defined by*:

$$f_{\sigma\sigma'}(\mathbf{q},\mathbf{p})=\int d^3s\, e^{-(i/\hbar)\mathbf{p}\cdot\mathbf{s}}\rho_{\sigma\sigma'}\left(\mathbf{q}+\frac{\mathbf{s}}{2},\mathbf{q}-\frac{\mathbf{s}}{2}\right). \quad \text{(D.49)}$$

Since ρ is a Hermitian operator, (D.49) must stay the same if we replace ρ by ρ^+. If, in addition, we take the complex conjugate, we see that $f_{\sigma\sigma'}(\mathbf{q},\mathbf{p})$ is a Hermitian matrix in the spin indices. The diagonal elements $f_{\sigma\sigma}=f_\sigma$ are therefore real functions of $\mathbf{q}$ and $\mathbf{p}$. If we neglect the spin orbit force, the Hamiltonian is invariant with respect to spin reversal, therefore $f_\sigma=f_{-\sigma}$. Under this condition the time-reversal properties of $f_\sigma(\mathbf{q},\mathbf{p},t)$ are very simple: f_σ is even (odd) in $\mathbf{p}$ according to whether f_σ is time even (odd).

The inverse transformation to (D.49) is given by

$$\rho_{\sigma\sigma'}(\mathbf{r},\mathbf{r}')=\frac{1}{(2\pi\hbar)^3}\int d^3p\, e^{(i/\hbar)\mathbf{p}(\mathbf{r}-\mathbf{r}')}f_{\sigma\sigma'}\left(\frac{\mathbf{r}+\mathbf{r}'}{2},\mathbf{p}\right). \quad \text{(D.50)}$$

Often, we wish to know the Wigner transform of products of operators like $\rho_W=(\exp(i\chi)\rho_0\exp(-i\chi))_W$. For this purpose we write the Wigner transform of the product of two general single-particle operators A and B in the following form.

$$(AB)_W=\int d^3r\, d^3r'\, d^3r''\, e^{-(i/\hbar)\mathbf{p}(\mathbf{r}-\mathbf{r}')}\delta\left(\mathbf{q}-\frac{\mathbf{r}+\mathbf{r}'}{2}\right)\langle\mathbf{r}|A|\mathbf{r}''\rangle\langle\mathbf{r}''|B|\mathbf{r}'\rangle. \quad \text{(D.51)}$$

The inverse Wigner transform (D.50) can also be written as

$$\langle\mathbf{r}|A|\mathbf{r}'\rangle=\frac{1}{(2\pi\hbar)^3}\int d^3p\, d^3q\, A(\mathbf{q},\mathbf{p})e^{(i/\hbar)\mathbf{p}(\mathbf{r}-\mathbf{r}')}\delta\left(\mathbf{q}-\frac{\mathbf{r}+\mathbf{r}'}{2}\right). \quad \text{(D.52)}$$

Inserting (D.52) into (D.51) and performing the $\mathbf{r},\mathbf{r}'$, and $\mathbf{r}''$ integrations gives:

$$(AB)_W=\frac{1}{(\pi\hbar)^6}\int d^3p'\, d^3p''\, d^3q'\, d^3q''\, e^{2i(\mathbf{p}''\mathbf{q}'-\mathbf{p}'\mathbf{q}'')/\hbar}\times A(\mathbf{q}+\mathbf{q}',\mathbf{p}+\mathbf{p}')B(\mathbf{q}+\mathbf{q}'',\mathbf{p}+\mathbf{p}''). \quad \text{(D.53)}$$

* In the following we use $\rho_{\sigma\sigma'}(\mathbf{q}+\mathbf{s}/2,\mathbf{q}-\mathbf{s}/2)=\rho(\mathbf{r},\sigma;\mathbf{r}',\sigma')$ with the spin indices σ,σ' and the center of mass coordinates $q=(\mathbf{r}+\mathbf{r}')/2$, $\mathbf{s}=\mathbf{r}-\mathbf{r}'$.

Replacing $A(\mathbf{q}+\mathbf{q}', \mathbf{p}+\mathbf{p}')$ and $B(\mathbf{q}+\mathbf{q}', \mathbf{p}+\mathbf{p}')$ in (D.53) by

$$\begin{pmatrix} A \\ B \end{pmatrix}(\mathbf{q}+\mathbf{q}', \mathbf{p}+\mathbf{p}') = e^{\mathbf{q}'\partial/\partial\mathbf{q}+\mathbf{p}'\partial/\partial\mathbf{p}}\begin{pmatrix} A \\ B \end{pmatrix}(\mathbf{q}, \mathbf{p}) \tag{D.54}$$

allows us to perform the $\mathbf{p}''$ and $\mathbf{q}''$ integrations, and the δ-functions allow us to do the $\mathbf{p}'$ and $\mathbf{q}'$ integration. After careful book-keeping of which derivatives act on which variables, we obtain [Gr 46, KK 76, Vo 77, BP 77b]:

$$(AB)_{\mathrm{W}} = A(\mathbf{q}, \mathbf{p}) e^{(i\hbar/2)\overleftrightarrow{\Lambda}} B(\mathbf{q}, \mathbf{p}) \tag{D.55}$$

with

$$\overleftrightarrow{\Lambda} = \overleftarrow{\nabla}_q \vec{\nabla}_p - \overleftarrow{\nabla}_p \vec{\nabla}_q, \tag{D.56}$$

where the direction of the arrows indicates whether the gradient acts to the left or to the right.

As a small example, let us calculate the Wigner transform of the kinetic energy density matrix (13.13):

$$\begin{aligned} \tau(\mathbf{q}, \mathbf{p}) &= \left(\frac{\vec{\partial}}{\partial \mathbf{r}} \rho(\mathbf{r}\mathbf{r}') \frac{\overleftarrow{\partial}}{\partial \mathbf{r}'} \right)_{\mathrm{W}} \\ &= -\frac{i}{\hbar} \mathbf{p} e^{(i\hbar/2)\overleftrightarrow{\Lambda}} f(\mathbf{q}, \mathbf{p}) e^{(i\hbar/2)\overleftrightarrow{\Lambda}} \frac{i}{\hbar} \mathbf{p} \\ &= \frac{p^2}{\hbar^2} \cdot f(\mathbf{q}, \mathbf{p}) + \frac{1}{4} \frac{\partial^2}{\partial \mathbf{q}^2} f(\mathbf{q}, \mathbf{p}). \end{aligned} \tag{D.57}$$

Integration over $\mathbf{p}$ gives the local part of the density matrix [see (D.50)]:

$$\frac{\hbar^2}{2m} \tau(\mathbf{q}) = \frac{1}{(2\pi\hbar)^3} \int d^3p \, \frac{p^2}{2m} f(\mathbf{q}, \mathbf{p}) + \frac{1}{4} \frac{\hbar^2}{2m} \Delta\rho(q). \tag{D.58}$$

In an actual calculation of the kinetic energy, the second term in (D.58) does not contribute because it is a divergence of a vector field.With formula (D.55) it is also very easy to calculate the Wigner transform of a commutator:

$$\frac{1}{\hbar}[A, B]_{\mathrm{W}} = \frac{2i}{\hbar} A(\mathbf{q}, \mathbf{p}) \cdot \sin\left(\frac{\hbar}{2} \overleftrightarrow{\Lambda} \right) \cdot B(\mathbf{q}, \mathbf{p}). \tag{D.59}$$

To lowest order in $\hbar$, the commutator just gives the classical Poisson bracket.